

Development of a Portable Electronic Nose Device for Sensing Harvested Fruit Volatile Organic Compounds

Lance Garcia
Department of Manufacturing
Engineering and Management
De La Salle University
Manila, Philippines
lance_garcia@dlsu.edu.ph

Raphael Antoine Lim
Department of Manufacturing
Engineering and Management
De La Salle University
Manila, Philippines
raphael_antoine_lim@dlsu.edu.ph

Nester Neil Lopez
Department of Manufacturing
Engineering and Management
De La Salle University
Manila, Philippines
nester_lopez@dlsu.edu.ph

Fernando Jr. Magallanes
Department of Manufacturing
Engineering and Management
De La Salle University
Manila, Philippines
fernando_magallanes@dlsu.edu.ph

Ronnie Concepcion
Department of Manufacturing
Engineering and Management
De La Salle University
Manila, Philippines
ronnie.concepcion@dlsu.edu.ph

Ryan Rhay Vicerra
Department of Manufacturing
Engineering and Management
De La Salle University
Manila, Philippines
ryan.vicerra@dlsu.edu.ph

Abstract—Climacteric fruits are considered a staple among many communities globally. However, wastage of these fruits due to overripe qualities continues to be a problem as fruits are commonly thrown out immediately. Since the traditional method of determining fruit quality remains subjective and often inconsistent, a more scientific and accurate means of determining fruit ripeness is needed. This study developed a portable electronic nose (e-nose) prototype containing calibrated gas, temperature and humidity sensors with a design enclosure that would house all electronic components. A geodesic dome was designed and fabricated for the enclosure using fused deposition modelling 3D printing. The design was considered due to its compressive strength, wind resistance, and air flow ventilation. Moreover, the portable e-nose was equipped with a sensor chamber consisting of a sensor array of MQ3, MQ4, MQ135, and DHT11 sensors. These would detect volatile organic compounds (VOCs) emitted by Lakatan banana while monitoring the temperature and humidity levels throughout the detection.

Keywords—digital agriculture, future food system, postharvest technology, product design, rapid prototyping, sustainable agriculture

I. INTRODUCTION

The utilization of electronic nose (e-nose,) for post-harvest crop quality classification specifically on ripeness and evaluation of crop quality has been used in recent years [1]. Climacteric crops have often been the primary crop target for this technology as they easily emit volatile organic compounds (VOCs). This is done in conjunction with the electronic nose to capture the surrounding air within proximity and utilize sensors to distinguish the VOCs within the captured air.

The quality of fruits change throughout their development. These changes are influenced by several factors, both internal and external. One of which is a fruit's ripening pattern [2-4]. Fruits classified as climacteric are said to be characterized by an increased rate of respiration that is often attributed to a high

discharge of ethylene [5]. The increased production of ethylene naturally accelerates the ripening of a given fruit. Therefore, climacteric fruits are deemed to cause significant problems regarding post-harvest storage, particularly during transport [6].

Electronic nose technology has shown great potential for multiple post-harvest crop detection and quality classification [7-9]. Further research is needed to optimize the technology and conduct a practical deployment of an electronic nose for use in the agricultural industry. Further consideration may be added for the study of quality classification based on ripeness and firmness as conducted by [10].

The general objective of this study is to develop a calibrated portable electronic nose system for the detection of emitted volatile organic compounds of Lakatan Bananas in terms of parts per million (ppm) levels. This study specifically aims to produce the following: (1) to design and fabricate e-nose, enclosure using additive manufacturing with geodesic dome geometry that will encase electronics subsystem; (2) to integrate calibrated gas, temperature, and humidity sensors as a functional portable sensor array for the detection of crop-emitted volatile organic compounds of Lakatan bananas.

This study will be limited to the manufacturing process of the e-nose thus, no verification of experimental data will be done. A Lakatan banana was used as the sample to verify that the system detects VOC concentration since the fruit is very common locally and that bananas exhibit relatively higher concentrations of VOCs [11]. Additionally, this proposed design will require a separate container wherein the fruit to be analyzed must be as close as possible to the entrance of the e-nose.

II. METHODOLOGY

As shown in Fig. 1, the conceptual framework of the proposed e-nose system indicates that the main input data will be in the form of the VOCs that diffuse from the fruit to be

analyzed. After which, the gas concentrations will be detected and processed by the sensor array and Arduino Mega microprocessor inside the e-nose, enclosure. Lastly, the output are the specific VOC concentration levels that are stored in an SD card for later analysis of gotten values.

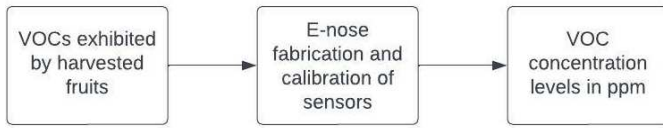


Fig. 1. Simplified structure of the concept flow of developing a portable electronic nose device for sensing harvested fruit volatile organic compounds.

A. Design of the e-nose, Enclosure and Sensor Chamber

The entire prototype was designed in Fusion 360 which is a Computer Aided Design (CAD) software developed by Autodesk. The software can export the 3D models as standard tessellation language (STL) files, which is the filetype supported by 3D printing slicer software during fabrication. Fig. 2 shows the first part of the enclosure in the Fusion 360 interface. The dome-like structure will completely cover the sensor chamber and electronics required for the system. It was determined that a diameter of 230 mm was required to fit all components including allowances for inaccuracies during the fabrication process. A geodesic shape was adapted into this design due to its compressive strength, uniform temperature and humidity distribution, wind resistance, and internal air flow ventilation properties. This way, the dome will be able to protect the sensitive components while maintaining proper airflow to avoid overheating the contents. Additionally, the dome has four extrusions near the bottom to serve as locking mechanisms when attached to the base.



Fig. 2. Geodesic dome enclosure design.

The second part of the enclosure is the base which will serve as the platform on which the electronics will rest and for air to flow into the sensor chamber as shown in Fig. 3. The diameter of the base is also 230 mm, and the perimeter matches that of the geodesic dome which will fit against the dome once locked in place. An opening was provided at the center of the base which is an 80 mm square with filleted corners of 35 mm radii. This opening serves as the entry point for the VOCs and a fan will be placed above leading into the sensor chamber.

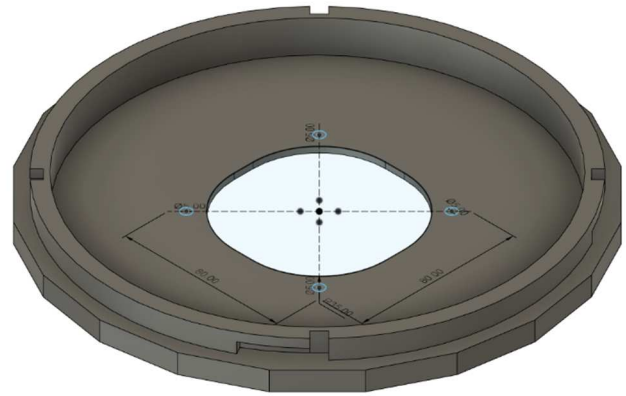


Fig. 3. Twist-on base design.

The sensor chamber is the crucial component of the e-nose, as it will house the sensors that will detect the VOCs. The design of the sensor chamber is based on the 80-mm size fan at the base with the sensors placed in the middle and chamfered 80 mm from the top to be aligned with the 5-mm hole at the top of the geodesic dome.

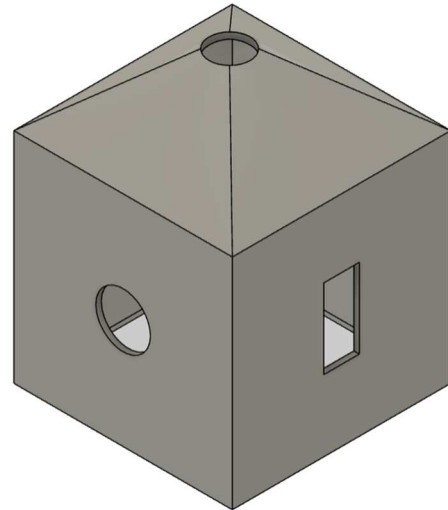


Fig. 4. Sensor chamber design.

To verify the good airflow within the design of the sensor chamber, the study conducted an airflow analysis using computational fluid dynamics (CFD) through Autodesk CFD where the result is shown in Fig. 5. Preparation was done through Fusion 360 fluid volume to define the volume where air can interact within the chamber. Boundary conditions are set with a velocity of 0.17 m/s on the inlet at the bottom of the chamber to mimic fan speed, a 0 MPa pressure gage on the outlet at the top of the chamber and lastly, a slip-symmetry condition to dictate that same conditions are applicable at other half of the chamber. The CFD show that there is significant backflow once it reaches the chamfered part of the chamber. This signified that utilizing fan for inhaust for VOCs is not ideal and given the proximity of the setup, it was more appropriate to let passive rising air with VOCs reach to the sensors placed in the middle. The CFD also verifies that design is able to securely trap air within the chamber and justifies the ideal placement of the sensors wherein air passes directly along the path of the sensors.

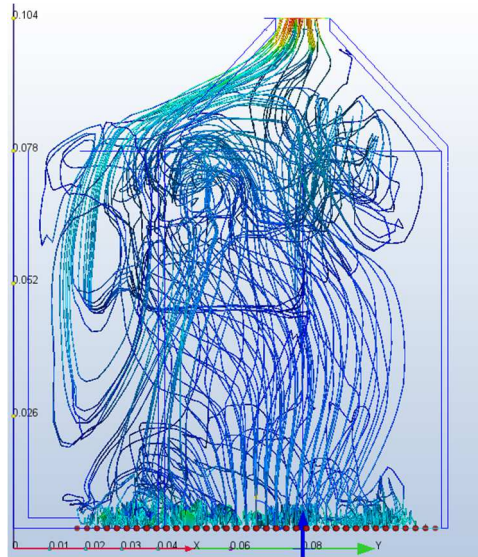


Fig. 5. CFD particle trace of airflow within sensor chamber.

B. Fabrication of the e-nose, Enclosure

The geodesic e-nose, enclosure was fabricated through 3D printing specifically fused deposition modelling (FDM) printing which is a form of additive manufacturing. For the material, polylactic acid (PLA) filament was used due to its accessibility and relatively high compressive strength compared to other 3D printing materials [12]. To cut down print time, the 3D model was printed separately in three parts namely, (1) the dome enclosure; (2) twist-on base; and (3) the sensor chamber. The entire prototype was printed on the Ultimaker S5 with the exception of the sensor chamber which was printed using third-party services, and the models were sliced using Cura which is a free software and the recommended slicer for most 3D printers.

Using Cura's default print settings under a normal quality produced a total print time of more than 1 day. Since this study had a time constraint of a maximum of 16h per print, several settings were refined to reduce the total print time. Some notable changes are as follows: layer height was increased to 0.35 millimeters since this dictates the number of layers of the total print. Increasing this will significantly reduce print time but sacrifices quality; overall print speed was set to 80 mm/s which was the fastest print speed that the Ultimaker S5 could print without compromising the quality. The inner wall speed was set at 40 mm/s since the quality inside the dome wasn't important for this study; support overhang angle was set at 70° and support density was 10%, these are the minimum required support structure setting that can hold up the dome. Any less would cause the top part of the dome to collapse. The full list of changed settings are shown in Fig. 6 with the changed settings in *italics*.

Before printing the final enclosure, several prototypes were printed at a smaller scale to analyze any discrepancies in the quality of the print. Shown in Fig. 7 are some of the notable versions before the final prototype. The base of the dome was printed with the same slicer settings and was printed in approximately 6 hours. However, the extrusions on the dome were not compatible with those in the base resulting in the twist-

type mechanism unable to lock. Thus, the extrusions were filed down to a smaller height to fit inside the slots of the base.

Setting	Profile	Current	Unit
Layer Height	0.35		mm
Initial Layer Height	0.2		mm
Generate Support	True		
Build Plate Adhesion Type	brim		
Enable Prime Tower	False		
Heat Up Speed	1.6		°C/s
Cool Down Speed	0.75		°C/s
Top/Bottom Thickness	1		mm
Infill Density	10		%
Infill Pattern	grid		
Print Speed	80		mm/s
Wall Speed	Calculated		mm/s
Outer Wall Speed	25		mm/s
Inner Wall Speed	40		mm/s
Top/Bottom Speed	Calculated		mm/s
Support Speed	70		mm/s
Retraction Prime Speed	Calculated		mm/s
Support Overhang Angle	70		°
Support Pattern	grid		
Support Density	10		%
Support Z Distance	0.5		mm
Brim Width	5		mm
Raft Extra Margin	3		mm
Raft Air Gap	0.25		mm
Enable Prime Tower	False		

Fig. 6. Cura slicer settings to obtain less than 16-h print time.

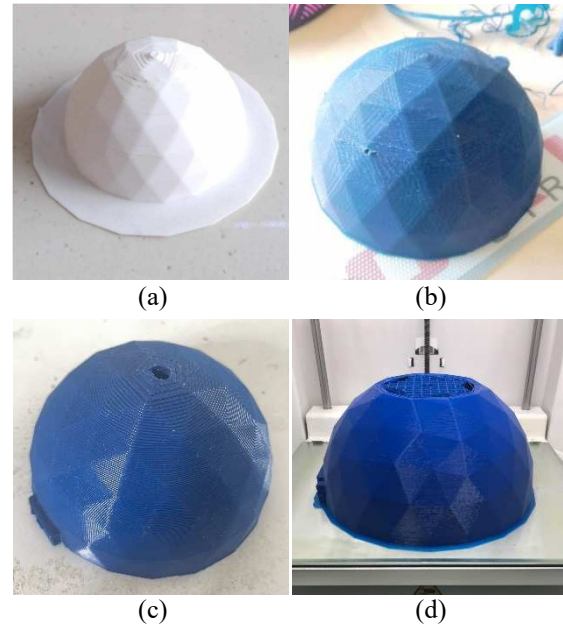


Fig. 7. (a) First 3D printed prototype at 10% scale, (b) 25% scale prototype, high print speeds produced a hole on the wall, (c) 25% scale prototype with display enclosure and exhaust, (d) full scale prototype partially printed.

C. Development of Sensing and Control Systems

Figure 8 is the visualization process of the interaction of each electronic components that is used in the study. Arduino Mega 2560 was used for it had many I2C pins suited for the sensors and devices connections. It was powered by a 5V DC battery. The input devices were the gas sensors (i.e., MQ3, MQ135, and MQ4), and the temperature and humidity sensor: DHT11, and the selection buttons. There were three buttons that controlled the system. First was the instant detection button, wherein it instructed the device to instantly classify the ripeness level of the banana. Secondly, the second button records the data continuously. Lastly, the third button terminates the recording from the second button. Then, the recorded data would be stored

in the SD Card Module. Whereas the instant detection would directly display its output through the OLED Display. The input and output devices were connected to each other through wires and the universal PCB.

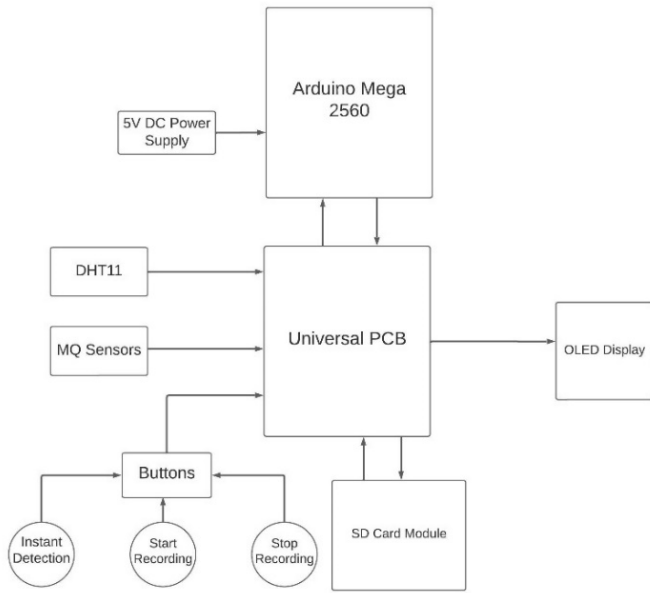


Fig. 8. Portable e-nose, system block diagram

Moreover, Fig. 9 evinced how the fan system worked. It was separated to the main system for it had its own power supply. It would be manually controlled using the switch and potentiometer. This would be activated upon the cleaning process of the sensor chamber. To push the remaining VOCs after testing through the exhaust.

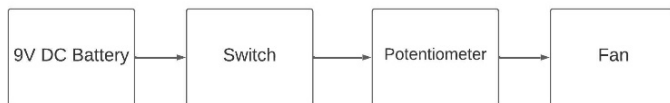


Fig. 9. Different versions of the prototype.

D. Modeling of the Algorithm

For the sensor chamber to function as an e-nose, the IF-THEN-ELSE conditional statements were selected to classify the maturity of Lakatan bananas based on the emitted VOCs measured in ppm. The ppm levels are obtained through the three MQ gas sensors embedded into the chamber. After which, the offset, derived from the one-point calibration, is added to the mean of the raw sensor values, which serve as the basis for classification. Only three maturity stages will be used to classify: Unripe, Ripe, and Overripe. The proposed system can only detect the presence of Lakatan bananas, and further experimentations are required to determine the respective ranges of the different stages.

The proposed e-nose offers two options for consumer usage, instant detection, and periodical data recording. The first feature provides users with the result of the classification within a minute. For 60 s, the sensor chamber will actively operate,

enabling the gas sensors to detect the VOCs every 5 s, resulting in 12 samples in total. The proposed classification algorithm will then be applied to determine the maturity of the Lakatan bananas. On the other hand, the second option stores the sensor values of the MQ gas sensors and DHT11 locally in the MicroSD card. Raw data is obtained every second until the third push button is manually pressed by the user, canceling the loop in the process.

III. RESULTS AND DISCUSSION

A. Complete Product of the Prototype

The final version of the e-nose prototype is shown in Fig. 10 where the universal PCB boards, electronic components and geodesic dome enclosure are visible. The PCB and other components are placed so that these will fit inside the geodesic dome. A 5-mm diameter rubber stub is used to plug the outlet at the top of the e-nose to trap the VOC-enriched air within the sensor chamber.

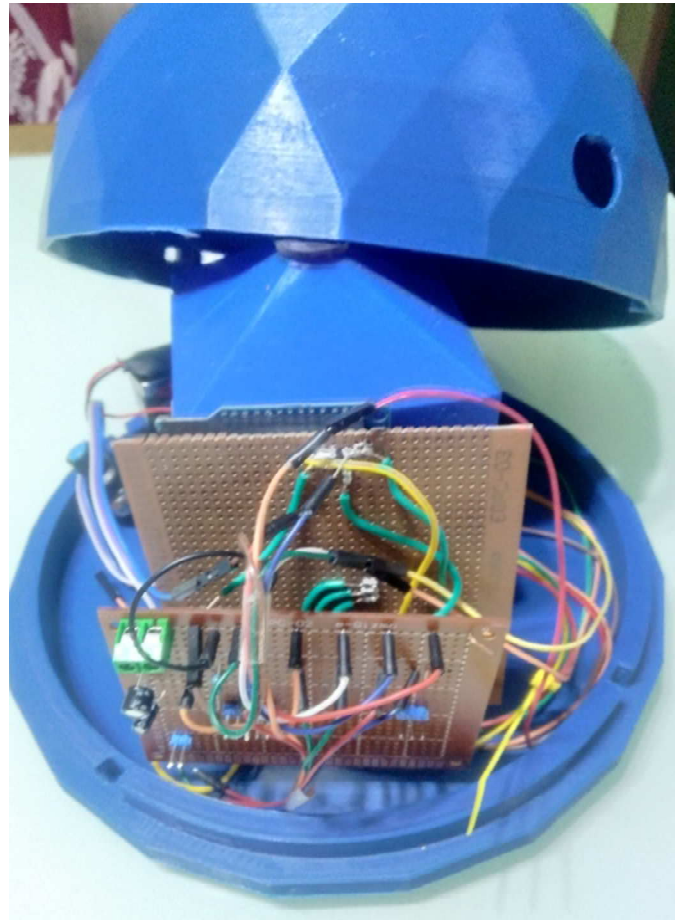


Fig. 10. Final version of e-nose prototype

Figure 11 shows the detailed wiring diagram that was soldered into the universal PCB board. Two boards were used, one that is connected directly to the Arduino Mega through pin headers while the other is mainly for all wirings to other electronic components. Additional notable components into the circuit were the addition of decoupling capacitors to ensure steady voltage that will power the whole circuitry.

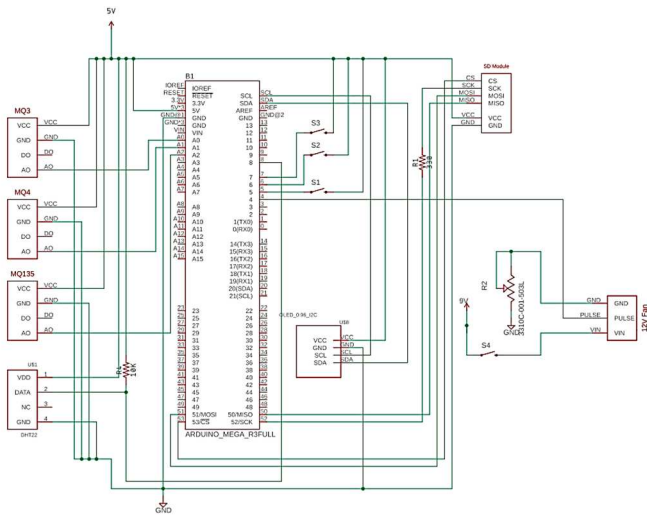


Fig. 11. Schematic of e-nose universal PCB

B. Experimentation and Data Gathering

During the experimentation process, it was conducted in a room with normal ambient temperature. The portable e-nose was placed above a 750 ml Tupperware containing two samples of Lakatan bananas. Then, after clicking the recording button, the sensors reading was collected within 10 min. Once the time was completed, it was then terminated using the stop recording button. The data was stored in the SD card module. After the data gathering, the sensor chamber of the e-nose, undergone the process of cleaning through the exhaust fan. This pushed the remaining VOCs outside the sensor chamber through the exhaust hole at the top of the e-nose. Moreover, the same process was repeated upon testing it without the banana sample.

C. Sensors Readings

Figure 13 displays the data recorded from the prototype. It was gathered from the SD card module, wherein it was converted to excel to graph the said data. With the presence of two samples of bananas, the ppm level of all the MQ sensors

increased gradually over time. On the other hand, without placing bananas, the sensor reading was flat. Thus, it can be concluded that the prototype could detect the presence of the banana. Humidity and temperature levels with banana are also monitored. The result reveals a constant temperature at 28°C and continuous increase of humidity with the samples. Whereas the humidity remained constant without the samples. This developed postharvest technology could be further enhanced by incorporating advanced computation techniques like Edge Computing embedded into the device [13-15] for faster predictions and mitigations. Moreover, it has great potential for adaptation in permaculture, agricultural supply chain, and space farming where VOCs are important to be monitored for management purposes [16-18]. Hence, this study given with the limitations of limited fruit, particularly banana, that can be sampled at a time, it provided a starting point for portable product design that would fit the basic needs of monitoring perishable climacteric fruits.



Fig. 12. Trial experiment using e-nose prototype.

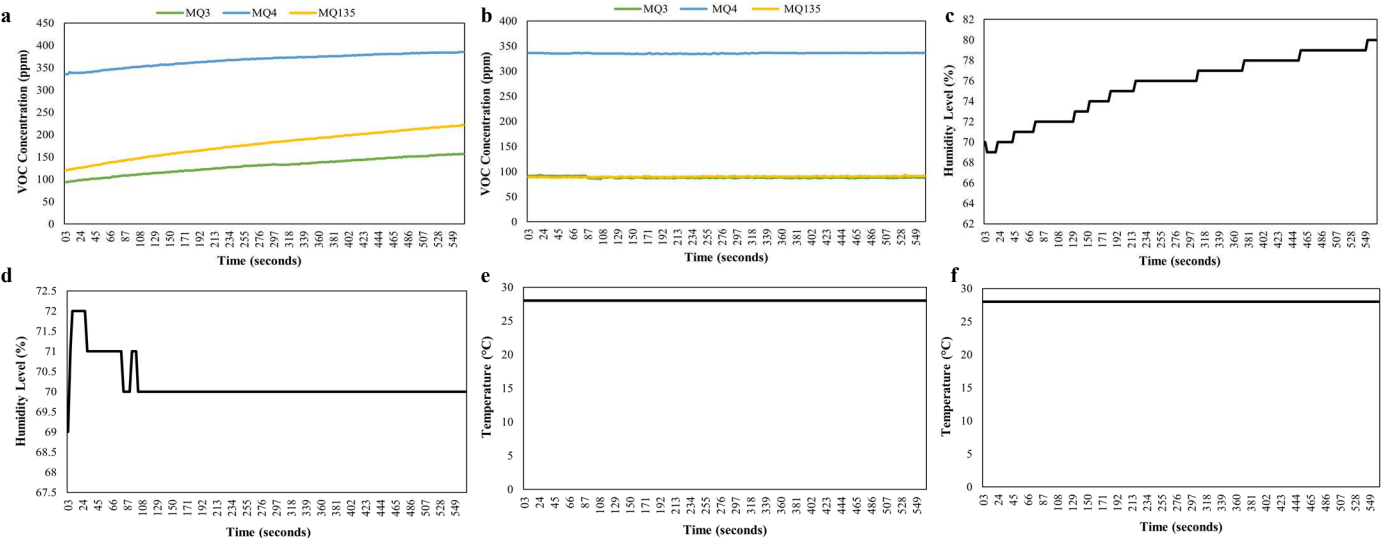


Fig. 13. Sensor readings with respect to time of (a) MQ3, MQ4, MQ135 with Lakatan bananas, (b) MQ3, MQ4, MQ135 without Lakatan bananas, (c) humidity level with Lakatan bananas, (d) humidity level without Lakatan bananas, (e) temperature levels with Lakatan bananas, and (f) temperature level without Lakatan bananas.

IV. CONCLUSION AND RECOMMENDATIONS

The study was successful in developing a working portable e-nose prototype through 3D-fabrication techniques and supplementary analyses to ensure proper detection of VOCs within the setup. Further recommendations to improve the prototype further are the following: Improve on the 3D printing approach for better quality print; Design a unique PCB to easily solder connections of components; Better space management around the base of the e-nose to fit with ease the different electronic components while ensuring it can be troubleshot; Design a better sensor chamber for better air concentration within sensor array; Integrate internet of things to send data directly to the cloud for ease to analysis of sensor data.

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